

Olist marketplace analysis — notebook and findings

Deliverable: `olist_marketplace_analysis.ipynb` (64 cells: 45 code, 19 markdown), plus an executed copy with all 27 charts rendered. Validated by running end-to-end against the real 9-file Olist dataset — 45/45 cells execute with no errors.

How to run it

Open in Colab, drop the nine CSVs into the session (or set `DATA_DIR` to a Drive folder), run all. The loader searches the local folder, then Drive, then falls back to a file-picker; it also checks each table's row count against the problem statement on load.

Structure

Setup → data understanding (key/referential-integrity/grain checks) → cleaning with a logged decision trail → feature engineering into one order-level master table → EDA baseline → Core Q1–Q6 → three deep dives (repeat customers, seller performance, anomalies) → Portuguese review-text mining → generated executive summary → appendix. Every technical block is followed by a plain-language box, and headline numbers are filed into a `FINDINGS` registry that generates the final summary, so the narrative can't drift from the data.

What the data actually shows (all figures from the executed run)

1. **Growth was volume, not basket size.** Orders grew 8.1x from Jan 2017 to Aug 2018; AOV flat; revenue tracks volume.
2. **Satisfaction did NOT decline as the platform scaled** — no significant time trend (Spearman $\rho = +0.34$, $p = 0.14$), ending higher than it started. It *swings*, and the swings track the monthly late-delivery rate ($\rho = -0.48$, $p = 0.031$), not volume ($\rho = -0.08$, $p = 0.75$). Worst months (Mar 2018: 3.73 stars

/ 21% late; Feb 2018; Nov 2017 Black Friday) are the worst delivery months, each followed by recovery.

3. **Lateness is the primary driver of dissatisfaction.** Late orders: 54.6% rated 1–2★ vs 9.5% on time (RR 5.75, CI 5.59–5.92; Cliff's $\delta = -0.55$; Cramér's $V = 0.37$). Clean dose–response. Late orders are 8.1% of deliveries but 33.7% of all bad reviews.
4. **The broken promise matters more than the wait** (2×2 , holding each constant): being late multiplies the 1–2★ rate 2.0x (short waits) / 5.1x (long waits); doubling the wait while staying on time multiplies it only 1.4x. Estimated-delivery-date logic is itself a lever.
5. **Geography and category are largely lateness/slowness in disguise.** 59.7% of sellers in SP; distance → delivery days $\rho = +0.54$, → freight $\rho = +0.58$. Restricting to on-time orders shrinks the regional 1–2★ spread from 5.1% to 1.6% and the category spread from 1.32 to 0.89 stars (–32%). Weakest category is "unknown" (missing catalogue data), 3.14 stars.
6. **Payment explains spend, not satisfaction.** Installments vs order value $\rho = +0.37$; payment type vs review $\eta^2 = 0.0001$, and the installment effect vanishes within order-value quartiles.
7. **Root cause, decomposed** (mutually exclusive, sums to 100%): on-time & fast 51.5% of bad reviews · >7 days late 17.5% · never delivered 15.4% (5.2x over-represented) · 0–7 days late 11.0% · on-time but very slow 4.5%. So **48.5% of bad reviews are delivery-related**; the residual 51.5% is a product/listing problem.
8. **Sellers differ, and the difference is punctuality.** Between-seller $\eta^2 = 0.03$; after empirical-Bayes shrinkage and controlling for category/destination mix, residual spread p10 → p90 = –0.24 to +0.26 stars; seller late rate vs shrunk score $\rho = -0.46$. 27 sellers are late significantly more often than chance (binomial + BH-FDR); the bottom decile is only 8.8% of revenue.
9. **Retention is a delivery outcome.** Only 3.12% of customers ever reorder. A late first delivery is associated with a 22% lower chance of a second order (3.78% → 2.94%; RR 0.78, CI 0.67–0.90). Notably the bad *score* is a weaker predictor than the late *delivery*.

10. **Review text corroborates independently.** 41% of negative comments use delivery language; top negative terms are `decepcao`, `devolucao dinheiro`, `descaso`, `enganosa`, `nao recebi`. Among bad reviews on fast on-time orders, delivery language falls to 22% and product-quality language rises to 33% — confirming the residual is a product problem. Asymmetry worth noting: positive reviews praise the *product* first ("otimo produto", "otima qualidade"); good delivery avoids punishment rather than earning praise.

Recommended priorities (in the notebook)

P1 manage the promise, not just the parcel · P2 attack non-delivery and the >7-day-late tail · P3 enforce seller fulfilment SLAs · P4 shorten distance on the worst lanes · P5 listing/packaging accuracy for the residual · P6 track retention by first-order delivery outcome.

Methodology notes

Effect sizes reported alongside every p-value (Cliff's δ , Cramér's V , η^2 , risk ratios with CIs); Wilson intervals on proportions; BH-FDR correction wherever many categories or sellers are tested; deconfounding by re-running the driver screen inside on-time deliveries only; PAF used to combine effect strength with prevalence. Association not causation is stated throughout, with limitations listed in the appendix